## Interactive / complex / 14

IC 1
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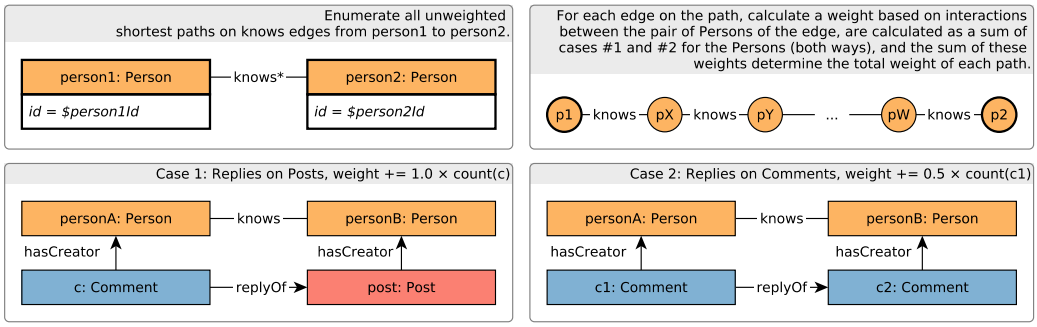
|---|---|
| query | Interactive / complex / 14 |
| title | Trusted connection paths |
| pattern |  |
| desc. | Given two Persons, find all (unweighted) shortest paths between these two Persons, in the subgraph induced by the knows relationship.<br>Then, for each path calculate a weight. The nodes in the path are Persons, and the weight of a path is the sum of weights between every pair of consecutive Person nodes in the path.<br>The weight for a pair of Persons is calculated based on their interactions:<br>• Every direct reply (by one of the Persons) to a Post (by the other Person) contributes 1.0.<br>• Every direct reply (by one of the Persons) to a Comment (by the other Person) contributes 0.5.<br>Note that interactions are counted both ways (e.g. if Alice writes 2 Post replies and 1 Comment reply to Bob, while Bob writes 3 Post replies and 4 Comment replies to Alice, their interaction score is $2 \times 1.0 + 1 \times 0.5 + 3 \times 1.0 + 4 \times 0.5 = 7.5$).<br>Return all the paths with shortest length, and their weights. Do not return any rows if there is no path between the two Persons. |
| params | 1 person1Id ID<br>2 person2Id ID |
| result | 1 personIdsInPath [ID] C identifiers representing an ordered sequence of the Persons in the path<br>2 pathWeight 64-bit Float C |
| sort | 1 pathWeight ↓ The order of paths with the same weight is unspecified |
| CPs | 3.3, 5.3, 7.2, 7.3, 7.5, 7.7, 8.1, 8.2, 8.3, 8.6 |
| relevance | This query looks for a variable length path, starting at a given Person and finishing at an another given Person. This is a more complex query as it not only requires computing the path length, but returning it and computing a weight. To compute this weight one must look for smaller sub-queries with paths of length three, formed by the two Persons at each step, a Post and a Comment. |